Informatica® Cloud Application Integration
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Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI

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Table of Contents

Preface 5

Chapter 1: Introduction to Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe 6

Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe contents . . 6

Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe assets . . . 7

Chapter 2: Using the Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe 8

Step 1. Copy and access the recipe . . . 8

Step 2. Configure and publish the Databricks Mosaic AI connection . . . 9

Step 3. Configure and publish the processes . . . 9

Step 4. Publish and run the guide . . . 10

Preface

Use *Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI* to learn how to extract structured JSON data from unstructured text. This recipe leverages the Databricks Mosaic AI Large Language Model (LLM) to accurately identify and organize relevant information. Automating this data transformation simplifies the analysis and integration of unstructured data into structured workflows.

CHAPTER 1

Introduction to Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe

The Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe is based on REST and SOAP APIs. Use this recipe to extract structured JSON data from unstructured text. This recipe leverages the Databricks Mosaic AI Large Language Model (LLM) to accurately identify and organize relevant information. Automating this data transformation simplifies the analysis and integration of unstructured data into structured workflows.

The recipe allows you to upload a file with context, provide field names as a user prompt to extract data, read the text from the file, extract the required fields, and convert them into JSON format.

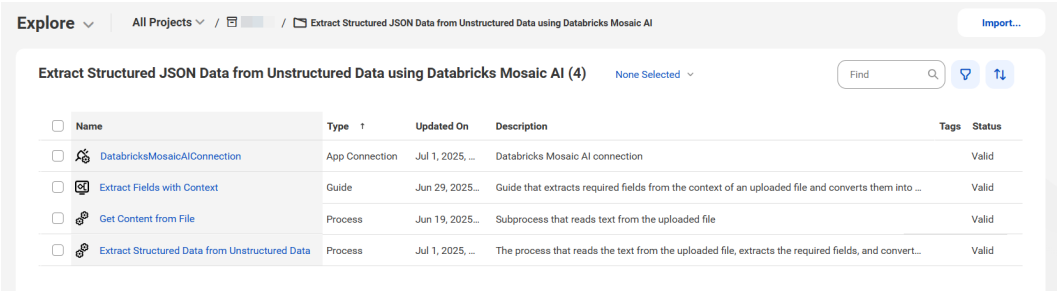
You can upload files of the following formats:

`.txt`, `.xml`, `.json`, `.html`, `.csv`, `.md`, and `.yaml`

Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe contents

The Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe contains an app connection, processes, and a guide.

The following image shows the assets that the Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe package contains:



Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe assets

The following table lists the assets that the Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe package contains:

Asset Name	Asset Type	Description
DatabricksMosaicAIConnection	App Connection	Databricks Mosaic AI connection.
Get Content from File	Process	Subprocess that reads text from the uploaded file.
Extract Structured Data from Unstructured Data	Process	The process that reads the text from the uploaded file, extracts the required fields, and converts them into JSON format.
Extract Fields with Context	Guide	Guide that extracts the required fields from the context of an uploaded file and converts them into structured data.

CHAPTER 2

Using the Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe

To use the Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Configure and publish the Databricks Mosaic AI connection.
3. Configure and publish the processes.
4. Publish and run the guide.

Step 1. Copy and access the recipe

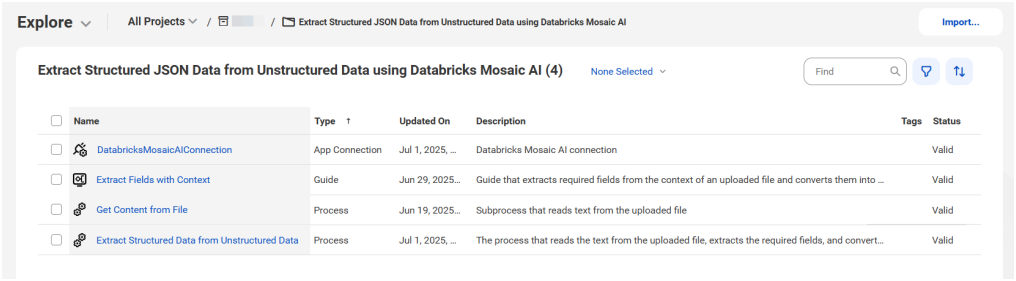
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **Extract Structured JSON Data from Unstructured Data using Databricks Mosaic AI** recipe and click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Configure and publish the Databricks Mosaic AI connection

Configure the host URL and access token to access the Databricks Mosaic AI instance, and then publish the DatabricksMosaicAIConnection connection.

1. Open the **DatabricksMosaicAIConnection** connection.
2. From the **Type** list, select **Databricks Mosaic AI**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Host	Host URL for Databricks Mosaic AI.
Bearer_Token	Access token for Databricks Mosaic AI.

5. Save and publish the connection.

Step 3. Configure and publish the processes

Configure the name of the serving endpoint in Databricks Mosaic AI and publish the processes.

1. To publish the **Get Content from File** process, click **Actions** in the row that contains the process and select **Publish**.
2. Open the **Extract Structured Data from Unstructured Data** process.
3. On the **Temp Fields** tab of the **Start** step, enter the name of the serving endpoint in Databricks Mosaic AI in the **ServingEndPointName** field. Default is **databricks-meta-llama-3-1-8b-instruct**. You can optionally edit the serving endpoint name.
4. Save and publish the process.

Step 4. Publish and run the guide

You can enter field names based on the context of an uploaded file to extract data.

1. Open the **Extract Fields with Context** guide.
2. On the **Start** tab of the Start step, ensure that the **Run As** field is set to **Current User**.
3. Save and publish the guide.
4. On the **Actions** menu, click **Run**. Alternatively, you can copy the execution URL from the **Properties Details** dialog box to run the guide.
5. On the **Instructions** page, select the context file that you want to upload from your machine.
6. In the **Enter field names to extract data** field, enter comma separated field names to extract data based on the file contents.
7. Click **Continue**.
   You will receive a response from the LLM.
8. You can extract new data with the same context or upload a new file to start a new conversation.
   - To extract new data, click **Next Question**. Enter new field names with the same context.
   - To upload a new file, click **New File**. Upload a new file from your machine and enter field names based on the file contents.
9. Click **Continue**.
   You will receive a response from the LLM.

Note: If the uploaded file is empty or the field names are not related to the file context, an error occurs.

You can also use the embed code to embed the guide into an HTML document of a third-party application.